

pattern_push_heterogeneous

January 3, 2026

```
[ ]: import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from pathlib import Path
import sys

sys.path.insert(0, str(Path(__file__).parent.parent))
from utils import (generate_patterns, setup_write_experiment,
    ↪setup_sleep_experiment,
                    run_cpp, read_parameters, DATA_DIR)

PLOTS_DIR = Path(__file__).parent.parent / "plots"
PLOTS_DIR.mkdir(exist_ok=True)
```

```
[ ]: sns.set_style("darkgrid")
sns.set_context("paper", font_scale=1.5)
plt.rcParams['text.usetex'] = True
plt.rcParams['font.family'] = 'serif'
plt.rcParams['font.serif'] = ['Times']
plt.rcParams.update({
    'font.size': 20,
    'axes.labelsize': 30,
    'axes.titlesize': 30,
    'xtick.labelsize': 25,
    'ytick.labelsize': 25,
    'legend.fontsize': 20,
    'figure.titlesize': 20,
    'lines.linewidth': 2.5,
    'axes.linewidth': 1.5,
    'axes.grid': False,
    'font.weight': 'bold'
})
```

```
[ ]: sparsities = [0.40, 0.45, 0.50, 0.55, 0.60]
N = 600
M = len(sparsities)
```

```

# Generate each pattern with its specific sparsity (sparsity = fraction of
↳ inactive units, P(0) convention)
# Matches generate_patterns_new: each bit is independently 1 with probability
↳ (1-s)
np.random.seed(42)
X_list = []
for s in sparsities:
    pattern = (np.random.rand(N) > s) # True 1, with probability (1-s)
    X_list.append(pattern)
patterns = np.array(X_list)

```

```

[ ]: config = setup_write_experiment(
    name="pattern_push_heterogeneous",
    patterns=patterns,
    params={"leak": 1.0, "drive_target": 6.0, "learning_rate": 0.0001,
           "momentum": 0.9, "max_iterations": 5000, "convergence_threshold": 0.
↳ 01}
)
run_cpp("write", config)

```

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous/config.json

Loaded 5 patterns of size 600

Running 1 training simulations

Training sim 0

Sim 0 converged in 97 iterations

All simulation data has been written to all_simulation_data.csv

Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous

```

[ ]: CompletedProcess(args=['/mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write', '/mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous/config.json'], returncode=0)

```

```

[ ]: sim_dir = DATA_DIR / "trained_networks" / "pattern_push_heterogeneous" /
↳ "sim_nb_0"
W = np.loadtxt(sim_dir / "weights.data")
X = np.loadtxt(sim_dir / "patterns.data")
v_N = np.ones(N) * 0.5

# Network derivative at neutral state
sigma = lambda x: 1.0 / (1.0 + np.exp(-x))

```

```

f = -v_N + sigma(W @ v_N)

# Compute push P_mu for each pattern
P = np.array([
    np.dot(f, (X[mu] - v_N) / np.linalg.norm(X[mu] - v_N))
    for mu in range(M)
])

print("Push values at neutral state (heterogeneous sparsities):")
for mu in range(M):
    print(f"  Pattern {mu} (s={sparsities[mu]:.2f}): P_{mu} = {P[mu]:.4f}")

```

Push values at neutral state (heterogeneous sparsities):

```

Pattern 0 (s=0.40): P_0 = 4.1754
Pattern 1 (s=0.45): P_1 = 3.7924
Pattern 2 (s=0.50): P_2 = 3.2664
Pattern 3 (s=0.55): P_3 = 3.0122
Pattern 4 (s=0.60): P_4 = 2.5921

```

```

[ ]: fig, ax = plt.subplots(figsize=(8, 4))

bars = ax.bar(range(M), P, facecolor='white', edgecolor='black', linewidth=2.5,
    width=0.6)
ax.axhline(0, color='black', linewidth=1.5)

# Add sparsity annotations above each bar
for mu, (bar, s) in enumerate(zip(bars, sparsities)):
    ax.annotate(f'$s$={s:.2f}', xy=(bar.get_x() + bar.get_width()/2, bar.
    get_height()),
        xytext=(0, 5), textcoords='offset points', ha='center',
        va='bottom', fontsize=25)

# Expand y-axis to fit annotations
ymin, ymax = ax.get_ylim()
ax.set_ylim(ymin, ymax + 0.15 * (ymax - ymin))

ax.set_xlabel(r'Pattern index $\mu$')
ax.set_ylabel(r'Push $P_\mu$')
ax.set_xticks(range(M))
ax.tick_params(axis='both', which='both', bottom=True, left=True, top=False,
    right=False)

for spine in ['bottom', 'left']:
    ax.spines[spine].set_visible(True)
    ax.spines[spine].set_color('black')
    ax.spines[spine].set_linewidth(1.5)
for spine in ['top', 'right']:

```

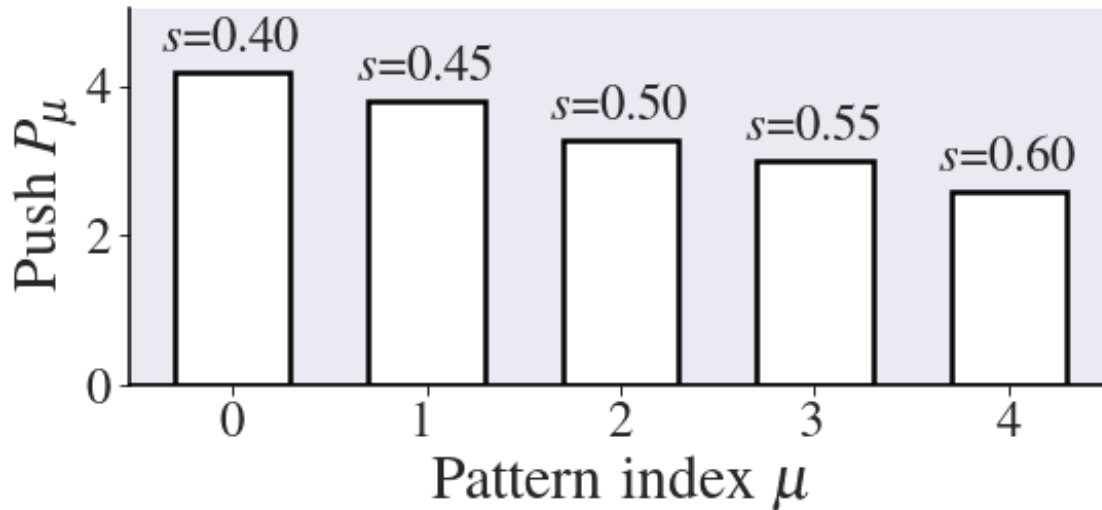
```

ax.spines[spine].set_visible(False)

plt.tight_layout()
plt.savefig(PLOTS_DIR / 'pattern_push_heterogeneous.png', dpi=300,
            bbox_inches='tight')
print(f"Saved to {PLOTS_DIR / 'pattern_push_heterogeneous.png'}")
plt.show()

```

Saved to /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/scripts/plots/pattern_push_heterogeneous.png



```

[ ]: sleep_config = setup_sleep_experiment(
    name="pattern_push_heterogeneous",
    trained_networks_dir=DATA_DIR / "trained_networks" /
    "pattern_push_heterogeneous",
    params={
        "beta": 0.1, "delta": 0.01, "max_queries": M,
        "stop_on_spurious": 0, "stop_on_all_found": 0,
        "save_inhibition_matrices": 1
    }
)
run_cpp("sleep", sleep_config)

```

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 4/5 patterns, 0 spurious

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous

```
[ ]: CompletedProcess(args=['/mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep', '/mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous/config.json'], returncode=0)
```

```
[ ]: sleep_dir = DATA_DIR / "sleep_results" / "pattern_push_heterogeneous" /   
    ↪ "sim_nb_0"  
results = pd.read_csv(sleep_dir / "results.data")  
recovered_indices = results["recovered_pattern_idx"].values  
  
num_iters = len(results)  
P_evolution = np.zeros((num_iters, M))  
  
for t in range(num_iters):  
    A = np.loadtxt(sleep_dir / f"inhib_matrix_iter_{t}.data")  
    A_diag = np.diag(A)  
    f_t = -v_N + sigma(W @ v_N) - A_diag * v_N  
    P_evolution[t] = [np.dot(f_t, (X[mu] - v_N) / np.linalg.norm(X[mu] - v_N))   
    ↪ for mu in range(M)]
```

```
[ ]: fig, axes = plt.subplots(num_iters, 1, figsize=(6, 2.5*num_iters), sharex=True,   
    ↪ sharey=True)  
axes = np.atleast_1d(axes)  
  
explored_set = set()  
for t in range(num_iters):  
    ax = axes[t]  
    colors = ['white'] * M  
  
    for idx in explored_set:  
        colors[idx] = '#FFD699' # Light orange (past explored)  
  
    if recovered_indices[t] >= 0:  
        colors[recovered_indices[t]] = '#d62728' # Red (just retrieved)  
        explored_set.add(recovered_indices[t])  
  
    bars = ax.bar(range(M), P_evolution[t], color=colors, edgecolor='black',   
    ↪ linewidth=2, width=0.6)  
    ax.axhline(0, color='black', linewidth=1)
```

```

ax.set_ylabel(f'Iter {t}')

# Add sparsity annotations above each bar (only on first subplot)
if t == 0:
    for mu, (bar, s) in enumerate(zip(bars, sparsities)):
        ax.annotate(f'$s$={s:.2f}', xy=(bar.get_x() + bar.get_width()/2,
→bar.get_height()),
                    xytext=(0, 5), textcoords='offset points', ha='center',
→va='bottom')

    for spine in ['bottom', 'left']:
        ax.spines[spine].set_visible(True)
        ax.spines[spine].set_color('black')
        ax.spines[spine].set_linewidth(1.5)
    for spine in ['top', 'right']:
        ax.spines[spine].set_visible(False)

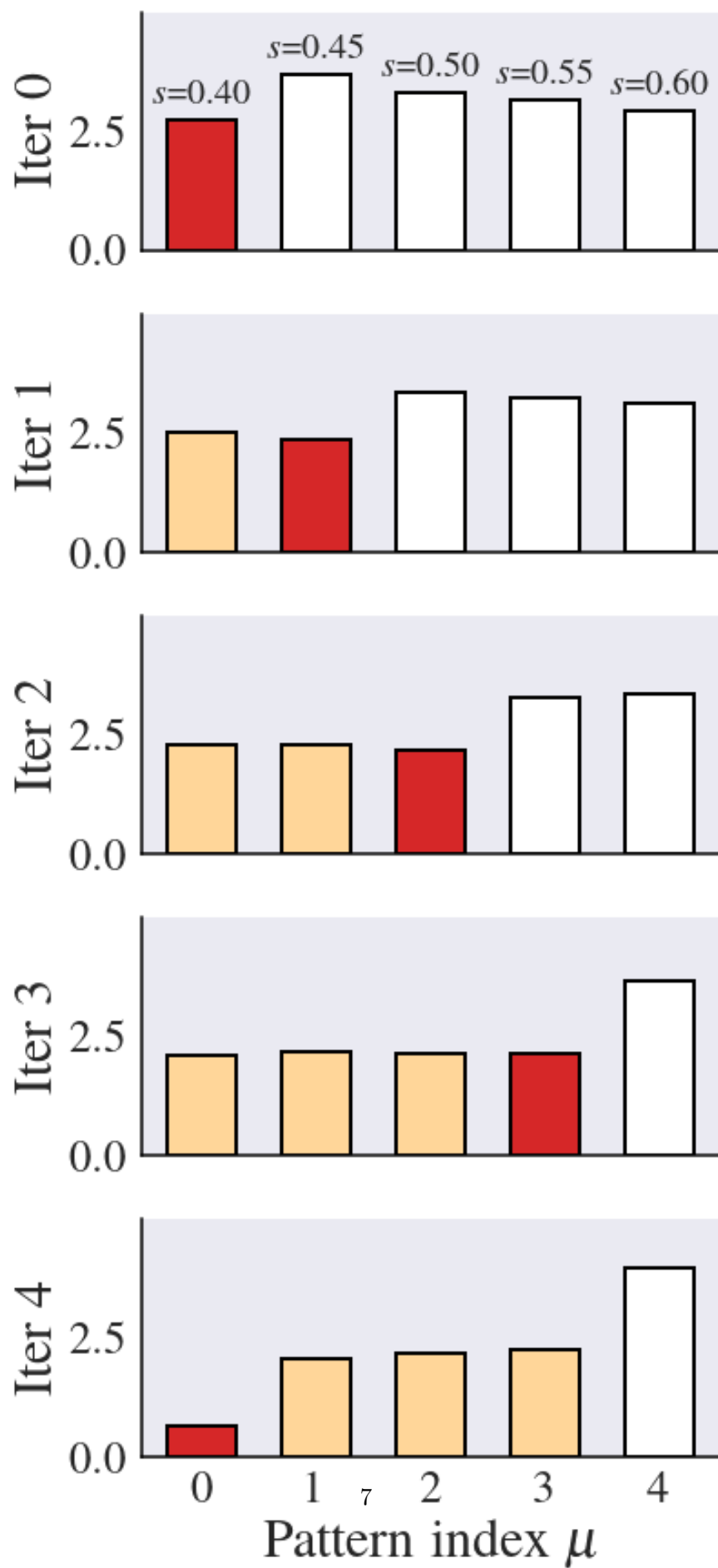
axes[-1].set_xlabel(r'Pattern index $\mu$')
axes[-1].set_xticks(range(M))

# Expand y-axis to fit annotations on first subplot
ymin, ymax = axes[0].get_ylim()
axes[0].set_ylim(ymin, ymax + 0.2 * (ymax - ymin))

plt.tight_layout()
plt.savefig(PLOTS_DIR / 'pattern_push_heterogeneous_AR_evolution.png', dpi=300,
→bbox_inches='tight')
print(f"Saved to {PLOTS_DIR / 'pattern_push_heterogeneous_AR_evolution.png'}")
plt.show()

```

Saved to /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/scripts/plots/pattern_push_heterogeneous_AR_evolution.png



Statistical Analysis: Visit Count and First Visit Order vs Sparsity

This section runs multiple independent simulations to analyze: 1. **Visit count**: How many times each pattern is retrieved based on its sparsity 2. **First visit order**: When each pattern is first discovered based on its sparsity

We use box plots to visualize the distribution across 30 independent simulations for each load condition ($p \in \{4, 6, 8\}$).

```
[ ]: from utils import list_simulations, build

LOADS = [4, 6, 8]
NB_REPETITIONS = 20
SPARSITY_MIN = 0.3
SPARSITY_MAX = 0.7
MAX_QUERIES_STATS = 200
N = 200
EXPERIMENT_NAME_STATS = "pattern_push_heterogeneous_stats"

print("\n" + "="*70)
print("STATISTICAL ANALYSIS: VISIT COUNT AND FIRST VISIT ORDER")
print("="*70)
print(f"Network size: {N}")
print(f"Loads: {LOADS}")
print(f"Repetitions per load: {NB_REPETITIONS}")
print(f"Sparsity range: [{SPARSITY_MIN:.2f}, {SPARSITY_MAX:.2f}]")
print("="*70)
```

```
=====
STATISTICAL ANALYSIS: VISIT COUNT AND FIRST VISIT ORDER
=====

Network size: 200
Loads: [4, 6, 8]
Repetitions per load: 20
Sparsity range: [0.30, 0.70]
=====
```

```
[ ]: build()
```

```
[ ]: True
```

```
[ ]: print("\n" + "="*70)
print("TRAINING PHASE")
print("="*70)
```



```

for p in LOADS:
    print(f"\nTraining {NB_REPETITIONS} networks for p = {p}...")

    for rep in range(NB_REPETITIONS):
        # Generate heterogeneous patterns for this repetition
        np.random.seed(rep * 1000 + p * 100)
        sparsities_rep = np.linspace(SPARSITY_MIN, SPARSITY_MAX, p)
        patterns_rep = np.array([(np.random.rand(N) > s) for s in
↪sparsities_rep])

        config = setup_write_experiment(
            name=f"{EXPERIMENT_NAME_STATS}_p{p}_rep{rep}",
            patterns=patterns_rep,
            params={"leak": 1.0, "drive_target": 6.0, "learning_rate": 0.0001,
↪"momentum": 0.9, "max_iterations": 5000,
↪"convergence_threshold": 0.01}
        )
        run_cpp("write", config)

    print(f"Completed training for p = {p}")

```

```

=====
TRAINING PHASE
=====

```

```

Training 20 networks for p = 4...
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p4_rep0/config.json
Loaded 4 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 248 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p4_rep0
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p4_rep1/config.json
Loaded 4 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 219 iterations

```

All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep1
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep2/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 252 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep2
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep3/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 224 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep3
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep4/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 272 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep4
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep5/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 246 iterations

All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep5
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep6/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 273 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep6
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep7/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 243 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep7
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep8/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 295 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep8
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep9/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 333 iterations

All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep9
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep10/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 243 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep10
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep11/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 249 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep11
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep12/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 224 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep12
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep13/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 222 iterations

All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep13
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep14/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 252 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep14
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep15/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 317 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep15
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep16/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 236 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p4_rep16
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep17/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 212 iterations

All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep17
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep18/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 256 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep18
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep19/config.json
 Loaded 4 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 255 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p4_rep19
 Completed training for $p = 4$

Training 20 networks for $p = 6$...
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep0/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 292 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p6_rep0
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep1/config.json
 Loaded 6 patterns of size 200

Running 1 training simulations
Training sim 0
Sim 0 converged in 286 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p6_rep1
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep2/config.json
Loaded 6 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 294 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p6_rep2
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep3/config.json
Loaded 6 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 332 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p6_rep3
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep4/config.json
Loaded 6 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 273 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_heterogeneous_stats_p6_rep4
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep5/config.json
Loaded 6 patterns of size 200

Running 1 training simulations
 Training sim 0
 Sim 0 converged in 329 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep5
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep6/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 291 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep6
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep7/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 296 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep7
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep8/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 252 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep8
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep9/config.json
 Loaded 6 patterns of size 200

Running 1 training simulations
 Training sim 0
 Sim 0 converged in 300 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep9
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep10/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 322 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep10
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep11/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 336 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep11
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep12/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 323 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep12
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep13/config.json
 Loaded 6 patterns of size 200

Running 1 training simulations
 Training sim 0
 Sim 0 converged in 320 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep13
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep14/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 297 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep14
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep15/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 282 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep15
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep16/config.json
 Loaded 6 patterns of size 200
 Running 1 training simulations
 Training sim 0
 Sim 0 converged in 305 iterations
 All simulation data has been written to all_simulation_data.csv
 Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
 collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
 eterogeneous_stats_p6_rep16
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p6_rep17/config.json
 Loaded 6 patterns of size 200

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Running 1 training simulations
Training sim 0
Sim 0 converged in 358 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p6_rep17
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p6_rep18/config.json
Loaded 6 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 284 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p6_rep18
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p6_rep19/config.json
Loaded 6 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 285 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p6_rep19
Completed training for p = 6

Training 20 networks for p = 8..
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep0/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 344 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep0
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

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ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep1/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 316 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep1
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep2/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 296 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep2
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep3/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 395 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep3
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep4/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 338 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep4
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

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ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep5/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 344 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep5
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep6/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 314 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep6
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep7/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 342 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep7
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep8/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 310 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep8
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

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ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep9/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 362 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep9
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep10/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 329 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep10
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep11/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 346 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep11
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep12/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 344 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep12
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

```

```

ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep13/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 377 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep13
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep14/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 352 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep14
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep15/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 354 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep15
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep16/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 335 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep16
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

```

```

ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep17/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 330 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep17
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep18/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 324 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep18
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/write /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep19/config.json
Loaded 8 patterns of size 200
Running 1 training simulations
Training sim 0
Sim 0 converged in 369 iterations
All simulation data has been written to all_simulation_data.csv
Training complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re
collection_CHN/simulations/main_write_sleep/data/trained_networks/pattern_push_h
eterogeneous_stats_p8_rep19
Completed training for p = 8

```

```

[ ]: print("\n" + "="*70)
      print("SLEEP PHASE")
      print("="*70)

      for p in LOADS:
          print(f"\nRunning sleep for p = {p}...")

          for rep in range(NB_REPETITIONS):
              trained_dir = DATA_DIR / "trained_networks" /
              ↳ f"{EXPERIMENT_NAME_STATS}_p{p}_rep{rep}"

```



```

sleep_config = setup_sleep_experiment(
    name=f"{EXPERIMENT_NAME_STATS}_p{p}_rep{rep}",
    trained_networks_dir=trained_dir,
    params={
        "beta": 0.05, "delta": 0.01, "max_queries": MAX_QUERIES_STATS,
        "noise_dynamics": 1, "stddev_dynamics": 0.01,
        "stop_on_spurious": 1, "stop_on_all_found": 1,
    }
)
run_cpp("sleep", sleep_config)

print(f"Completed sleep for p = {p}")

```

===== SLEEP PHASE =====

Running sleep for p = 4...

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep0/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 23)

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep0

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep1/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 18)

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep1

```

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep2/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 23)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep2
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep3/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 21)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep3
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep4/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 19)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep4
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep5/config.json
Found 1 trained networks
Running 1 sleep simulations

```

Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 20)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep5
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep6/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 25)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep6
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep7/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 18)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep7
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep8/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 25)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto

nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p4_rep8
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep9/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 18)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p4_rep9
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep10/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 27)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p4_rep10
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep11/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 21)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p4_rep11
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p4_rep12/config.json

Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 23)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep12
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep13/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 24)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep13
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep14/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 18)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep14
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep15/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 20)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep15
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep16/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 20)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep16
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep17/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 19)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep17
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep18/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 20)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep18
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

```

ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p4_rep19/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 4/4 patterns, 0 spurious (all recovered before spurious at iter 18)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p4_rep19
Completed sleep for p = 4

```

Running sleep for p = 6...

```

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep0/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 5/6 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep0
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep1/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 28)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep1
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep2/config.json
Found 1 trained networks

```

Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 27)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep2
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep3/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 23)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep3
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep4/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 18)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep4
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep5/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 28)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep5

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep6/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 33)

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep6

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep7/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 21)

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep7

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep8/config.json

Found 1 trained networks

Running 1 sleep simulations

Sleep sim 0

Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 19)

Extracted Sim ID 0

All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)

Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep8

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen

```

eous_stats_p6_rep9/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 26)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p6_rep9
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p6_rep10/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 32)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p6_rep10
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p6_rep11/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 43)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p6_rep11
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p6_rep12/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 24)
Extracted Sim ID 0

```

All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep12
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep13/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 22)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep13
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep14/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 32)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep14
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p6_rep15/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 17)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p6_rep15
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat

ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re-
 collection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen-
 eous_stats_p6_rep16/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 21)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto-
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p6_rep16
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat-
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re-
 collection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen-
 eous_stats_p6_rep17/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 31)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto-
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p6_rep17
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat-
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re-
 collection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen-
 eous_stats_p6_rep18/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/6 patterns, 0 spurious (all recovered before spurious at iter 26)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto-
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p6_rep18
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat-
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Re-
 collection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen-
 eous_stats_p6_rep19/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0

Sim 0: found 4/6 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p6_rep19
 Completed sleep for p = 6

Running sleep for p = 8...
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep0/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 7/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p8_rep0
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep1/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 4/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p8_rep1
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep2/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 5/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv

Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep2
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep3/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 46)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep3
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep4/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 25)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep4
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep5/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/8 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep5
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R

```

ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep6/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 7/8 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p8_rep6
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep7/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 7/8 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p8_rep7
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep8/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 28)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p8_rep8
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
eous_stats_p8_rep9/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 7/8 patterns, 1 spurious

```

Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep9
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep10/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 3/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep10
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep11/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 7/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep11
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep12/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 27)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep12

Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep13/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 6/8 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep13
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep14/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 7/8 patterns, 1 spurious
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep14
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep15/config.json
Found 1 trained networks
Running 1 sleep simulations
Sleep sim 0
Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 31)
Extracted Sim ID 0
All simulation data has been written to all_simulation_data.csv
Final results have been written to final_results.csv (1 simulations)
Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_push_heterogeneous_stats_p8_rep15
Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogeneous_stats_p8_rep16/config.json
Found 1 trained networks
Running 1 sleep simulations

Sleep sim 0
 Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 33)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p8_rep16
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep17/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 26)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p8_rep17
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep18/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 8/8 patterns, 0 spurious (all recovered before spurious at iter 25)
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto
 nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
 push_heterogeneous_stats_p8_rep18
 Running: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulat
 ions/main_write_sleep/bin/sleep /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_R
 ecollection_CHN/simulations/main_write_sleep/data/configs/pattern_push_heterogen
 eous_stats_p8_rep19/config.json
 Found 1 trained networks
 Running 1 sleep simulations
 Sleep sim 0
 Sim 0: found 6/8 patterns, 1 spurious
 Extracted Sim ID 0
 All simulation data has been written to all_simulation_data.csv
 Final results have been written to final_results.csv (1 simulations)
 Sleep simulations complete. Results in: /mnt/c/Users/pauls/Desktop/Work/PhD/Auto

nomous_Recollection_CHN/simulations/main_write_sleep/data/sleep_results/pattern_
push_heterogeneous_stats_p8_rep19
Completed sleep for p = 8

```
[ ]: print("\n" + "="*70)
print("LOADING RESULTS")
print("="*70)

all_data = []

for p in LOADS:
    sparsities_p = np.linspace(SPARSITY_MIN, SPARSITY_MAX, p)

    for rep in range(NB_REPETITIONS):
        sleep_dir = DATA_DIR / "sleep_results" / _
        ↪f"{EXPERIMENT_NAME_STATS}_p{p}_rep{rep}" / "sim_nb_0"
        results_file = sleep_dir / "results.data"

        if not results_file.exists():
            print(f" Warning: No results for p={p}, rep={rep}")
            continue

        results_df = pd.read_csv(results_file)

        # Compute visit counts and first visit order
        visit_counts = {mu: 0 for mu in range(p)}
        first_visit = {mu: None for mu in range(p)}

        for iter_idx, row in results_df.iterrows():
            pattern_idx = int(row['recovered_pattern_idx'])
            if 0 <= pattern_idx < p:
                visit_counts[pattern_idx] += 1
                if first_visit[pattern_idx] is None:
                    first_visit[pattern_idx] = iter_idx

        # Store data for each pattern
        for mu in range(p):
            all_data.append({
                "load": p,
                "repetition": rep,
                "pattern_idx": mu,
                "sparsity": sparsities_p[mu],
                "visit_count": visit_counts[mu],
                "first_visit": first_visit[mu] if first_visit[mu] is not None _
            ↪else MAX_QUERIES_STATS + 1
            })
```

```
df_stats = pd.DataFrame(all_data)
print(f"\nTotal data points: {len(df_stats)}")
```

```
=====
LOADING RESULTS
=====
```

Total data points: 360

```
[ ]: print("\n" + "="*70)
      print("FIGURE: Visit Count vs Sparsity")
      print("="*70)

      n_bins = 5
      bin_edges = np.linspace(SPARSITY_MIN, SPARSITY_MAX, n_bins + 1)
      bin_centers = [(bin_edges[i] + bin_edges[i+1])/2 for i in range(n_bins)]
      bin_labels = [f"{c:.2f}" for c in bin_centers]

      df_stats["sparsity_bin"] = pd.cut(df_stats["sparsity"], bins=bin_edges,
      ↪ labels=bin_labels, include_lowest=True)

      fig, axes = plt.subplots(1, len(LOADS), figsize=(5*len(LOADS), 5.5),
      ↪ sharey=True)
      axes = np.atleast_1d(axes)

      for i, p in enumerate(LOADS):
          ax = axes[i]
          df_p = df_stats[df_stats["load"] == p].dropna(subset=["sparsity_bin"])

          box_data = []
          valid_labels = []
          for label in bin_labels:
              data = df_p[df_p["sparsity_bin"] == label]["visit_count"].values
              if len(data) > 0:
                  box_data.append(data)
                  valid_labels.append(label)

          if not box_data:
              continue

          bp = ax.boxplot(box_data, patch_artist=True, widths=0.6)

          for box in bp['boxes']:
              box.set_facecolor('white')
              box.set_edgecolor('black')
              box.set_linewidth(2)
```

```

for whisker in bp['whiskers']:
    whisker.set_color('black')
    whisker.set_linewidth(2)
for cap in bp['caps']:
    cap.set_color('black')
    cap.set_linewidth(2)
for median in bp['medians']:
    median.set_color('#d62728')
    median.set_linewidth(2.5)
for flier in bp['fliers']:
    flier.set_markerfacecolor('black')
    flier.set_markeredgecolor('black')
    flier.set_markersize(5)

ax.set_xticklabels(valid_labels, rotation=45)
ax.set_title(rf'{p} patterns')

if i == 0:
    ax.set_ylabel(r'Visit count')

for spine in ['bottom', 'left']:
    ax.spines[spine].set_visible(True)
    ax.spines[spine].set_color('black')
    ax.spines[spine].set_linewidth(1.5)
for spine in ['top', 'right']:
    ax.spines[spine].set_visible(False)

fig.supxlabel(r'Sparsity $s$', fontsize = 30)
plt.tight_layout()
output_path = PLOTS_DIR / 'pattern_push_heterogeneous_visit_count.png'
plt.savefig(output_path, dpi=300, bbox_inches='tight')
print(f"Saved: {output_path}")
plt.show()

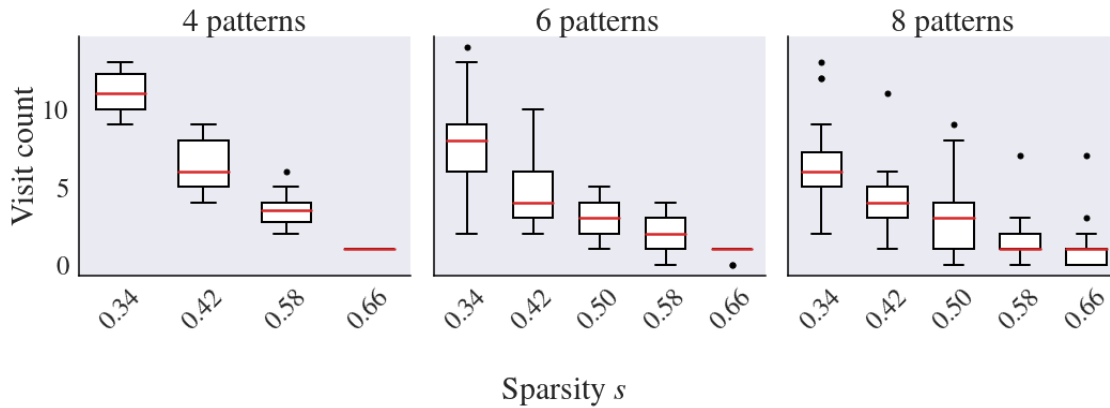
```

=====

FIGURE: Visit Count vs Sparsity

=====

Saved: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/scripts/plots/pattern_push_heterogeneous_visit_count.png



```
[ ]: print("\n" + "="*70)
print("FIGURE: First Visit Order vs Sparsity")
print("="*70)

df_visited = df_stats[df_stats["first_visit"] <= MAX_QUERIES_STATS].copy()
df_visited["sparsity_bin"] = pd.cut(df_visited["sparsity"], bins=bin_edges,
    ↳labels=bin_labels, include_lowest=True)

fig, axes = plt.subplots(1, len(LOADS), figsize=(5*len(LOADS), 5.5),
    ↳sharey=True)
axes = np.atleast_1d(axes)

for i, p in enumerate(LOADS):
    ax = axes[i]
    df_p = df_visited[df_visited["load"] == p].dropna(subset=["sparsity_bin"])

    box_data = []
    valid_labels = []
    for label in bin_labels:
        data = df_p[df_p["sparsity_bin"] == label]["first_visit"].values
        if len(data) > 0:
            box_data.append(data)
            valid_labels.append(label)

    if not box_data:
        continue

    bp = ax.boxplot(box_data, patch_artist=True, widths=0.6)

    for box in bp['boxes']:
        box.set_facecolor('white')
        box.set_edgecolor('black')
```

```

        box.set_linewidth(2)
    for whisker in bp['whiskers']:
        whisker.set_color('black')
        whisker.set_linewidth(2)
    for cap in bp['caps']:
        cap.set_color('black')
        cap.set_linewidth(2)
    for median in bp['medians']:
        median.set_color('#d62728')
        median.set_linewidth(2.5)
    for flier in bp['fliers']:
        flier.set_markerfacecolor('black')
        flier.set_markeredgecolor('black')
        flier.set_markersize(5)

    ax.set_xticklabels(valid_labels, rotation=45)
    ax.set_title(rf'{p} patterns')

    if i == 0:
        ax.set_ylabel(r'First visit iteration')

    for spine in ['bottom', 'left']:
        ax.spines[spine].set_visible(True)
        ax.spines[spine].set_color('black')
        ax.spines[spine].set_linewidth(1.5)
    for spine in ['top', 'right']:
        ax.spines[spine].set_visible(False)

fig.supxlabel(r'Sparsity $s$', fontsize = 30)
plt.tight_layout()
output_path = PLOTS_DIR / 'pattern_push_heterogeneous_first_visit.png'
plt.savefig(output_path, dpi=300, bbox_inches='tight')
print(f"Saved: {output_path}")
plt.show()

```

```

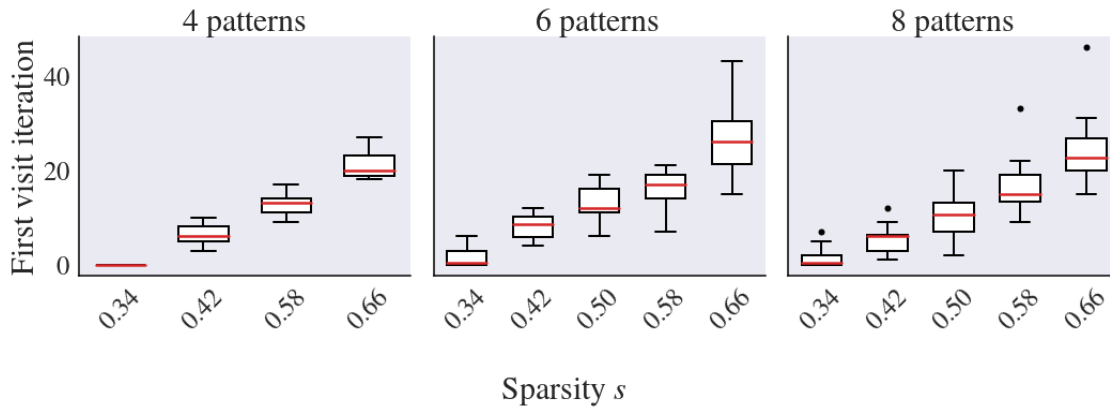
=====
FIGURE: First Visit Order vs Sparsity
=====

```

```

Saved: /mnt/c/Users/pauls/Desktop/Work/PhD/Autonomous_Recollection_CHN/simulations/main_write_sleep/scripts/plots/pattern_push_heterogeneous_first_visit.png

```



```
[ ]: print("\n" + "="*70)
print("SUMMARY STATISTICS")
print("="*70)

for p in LOADS:
    df_p = df_stats[df_stats["load"] == p]
    print(f"\n--- Load p = {p} ---")
    corr_visit = df_p["sparsity"].corr(df_p["visit_count"])
    print(f"Visit count - Sparsity correlation: r = {corr_visit:.3f}")

    df_p_visited = df_visited[df_visited["load"] == p]
    if len(df_p_visited) > 0:
        corr_first = df_p_visited["sparsity"].corr(df_p_visited["first_visit"])
        print(f"First visit - Sparsity correlation: r = {corr_first:.3f}")

print("\n" + "="*70)
print("ANALYSIS COMPLETE")
print("="*70)
```

```
=====
SUMMARY STATISTICS
=====

--- Load p = 4 ---
Visit count - Sparsity correlation: r = -0.943
First visit - Sparsity correlation: r = 0.966

--- Load p = 6 ---
Visit count - Sparsity correlation: r = -0.842
First visit - Sparsity correlation: r = 0.905

--- Load p = 8 ---
```


Visit count - Sparsity correlation: $r = -0.735$
First visit - Sparsity correlation: $r = 0.896$

=====
ANALYSIS COMPLETE
=====